

A BERT MODEL FOR PERSONAL DATA ANONYMIZATION

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This project reflects my work over the course of half a year as an AI engineer at Lawren.ai, where I developed my Python skills, learned Rust and discovered my true passion: Machine Learning. Having the opportunity to develop myself in this firm, my passion for Machine Learning grew and grew, and now it is an important part of my life. I was assigned in September with the task of handling data anonymization for the collaborators of Lawren.ai and this is my work.

This essay provides a detailed examination of the fine-tuning process applied to a BERT-based model for Named Entity Recognition (NER) in the Romanian legal domain. The project adapts the pre-trained `dumitrescustefan/bert-base-romanian-ner` model using the LegalNERo corpus, a manually annotated dataset containing 370 legal documents. This document traces the complete workflow from corpus preparation through model training, highlighting the technical decisions, preprocessing strategies, and training methodologies employed to achieve domain-specific NER capabilities.

Named Entity Recognition (NER) is a fundamental task in Natural Language Processing that involves identifying and classifying named entities in text, such as persons, organizations, locations, dates, and domain-specific entities. In the legal domain, accurate NER is crucial for document analysis, information extraction, and legal knowledge management systems.

This project addresses the challenge of adapting a general-purpose Romanian NER model to the specialized Romanian legal domain. The base model, `dumitrescustefan/bert-base-romanian-ner`, was pre-trained on general Romanian text (RONEC corpus) and recognizes standard entity types. However, legal documents contain specialized terminology, complex entity structures, and domain-specific references that require fine-tuning to achieve optimal performance. The reason for doing it is data anonymisation.

The fine-tuning process leverages transfer learning, where a pre-trained model's knowledge is adapted to a new task or domain through additional training on domain-specific data. This approach is particularly effective because BERT models have already learned rich linguistic representations during pre-training, requiring only task-specific adjustments rather than training from scratch. Through a hybrid approach leveraging both deep learning and pattern matching, the system achieves robust entity detection across multiple categories while ensuring consistent anonymization through intelligent deduplication and sequential numbering.

Corpus and Data Preparation

LegalNERo Corpus Structure

The LegalNERo corpus serves as the foundation for this fine-tuning project. It is a manually annotated corpus specifically designed for NER in the Romanian legal domain, containing 370 legal documents with gold-standard annotations. The corpus includes annotations for:

- **Persons (PER)**: Names of individuals mentioned in legal documents
- **Organizations (ORG)**: Legal entities, institutions, courts, and governmental bodies
- **Locations (LOC)**: Geographic locations, including cities, regions, and addresses
- **Time expressions (TIME)**: Dates, periods, and temporal references
- **Fine-grained Legal References (FGLEGAL)**: Specific legal document types including:
 - Law, Ordinance, Publication, Decree, Decision, Treaty, Report
 - Order, Regulation, Directive, Emergency Ordinance

- Norm, Convention, Code, and Other legal document types

Data Conversion Pipeline

The first critical step in the fine-tuning process involves converting the corpus from its native CoNLL-U Plus format to a format compatible with the Hugging Face Transformers library. The conversion script (`convert_conllup_to_json.py`) performs several essential operations:

****File Parsing**:** The script reads each `.conllup` file, parsing tab-separated values where:

- Column 1 contains token IDs
- Column 2 contains the token form (the actual word)
- Column 11 contains the NER annotation (`RELATE:NE`)

Sentence Segmentation: The script identifies sentence boundaries by detecting empty lines or comment lines (starting with `#`), grouping tokens into coherent sentences. This segmentation is crucial because NER models process text at the sentence level.

Data Structure Creation: Each sentence is converted into a JSON object containing:

```
{
  "tokens": ["Ion", "Popescu", "este", "procuror", ...],
  "ner_tags": ["B-PERSON", "I-PERSON", "O", "O", ...]
}
```

Split Management: The corpus includes a predefined split file (`splits_FGLEGAL_PER_LOC_ORG_TIME.tsv`) that divides the 370 documents into training, validation, and test sets. The conversion script respects these splits, ensuring that the model evaluation reflects true generalization performance.

Data Aggregation Strategy

The project implements a flexible data aggregation approach. The primary configuration (`USE_ALL_DATA_FOR_TRAINING = True`) combines all available data (train + validation + test) for training, then creates a **new 85%/15% split for training and validation**. This strategy maximizes the use of available annotated data, which is particularly valuable given the relatively small size of the corpus (370 documents). Alternatively, the script can preserve the original splits, using only the training set for model training and reserving validation and test sets for evaluation. This approach provides a more traditional evaluation setup but uses less data for training.

Model Selection and Architecture

Base Model: BERT for Romanian NER

The project uses `dumitrescustefan/bert-base-romanian-ner` as the base model. This model is built on BERT (Bidirectional Encoder Representations from Transformers) architecture, specifically adapted for Romanian language processing. Key characteristics include:

- Architecture: Transformer-based encoder with 12 layers, 768 hidden dimensions, and 12 attention heads
- Pre-training: Trained on large-scale Romanian text corpora
- Task Adaptation: Fine-tuned on RONEC (Romanian Named Entity Corpus) for general NER

- Vocabulary: Romanian-specific tokenizer with ~30,000 subword tokens

The choice of this model is strategic: it already understands Romanian language structure and general NER patterns, requiring only domain-specific adaptation rather than learning language fundamentals from scratch. It is pre-trained: learns general language patterns on large text (e.g., Wikipedia) and it can be fine-tuned (what I did) :adapts to specific tasks (classification, NER, Q&A).

Model Architecture for Token Classification

For the NER task, the base BERT model is extended with a classification head:

1. BERT Encoder: Processes input tokens and produces contextualized embeddings
2. Dropout Layer: Regularization to prevent overfitting
3. Linear Classification Layer: Maps each token's embedding to a probability distribution over entity labels

The classification head is initialized randomly and trained during fine-tuning, while the BERT encoder weights are updated with a lower learning rate to preserve general language knowledge while adapting to legal domain specifics.

BERT uses subword tokenization, which presents a critical challenge for NER: a single word may be split into multiple tokens. For example, the Romanian word "procuror" might be tokenized as `["proc", "##uror"]`. This creates an alignment problem: the original annotation assigns one label per word, but BERT produces multiple tokens per word.

Label Alignment Strategy

*Word-to-Token Mapping: The tokenizer's `word\_ids()` method provides a mapping from each token to its original word index

*Label Assignment:

- For the first subword token of a word: assign the word's original NER label
- For subsequent subword tokens: assign `-100` (a special value that tells the model to ignore this token during loss calculation)
- For special tokens (CLS, SEP, PAD): assign `-100`

*Example Alignment:

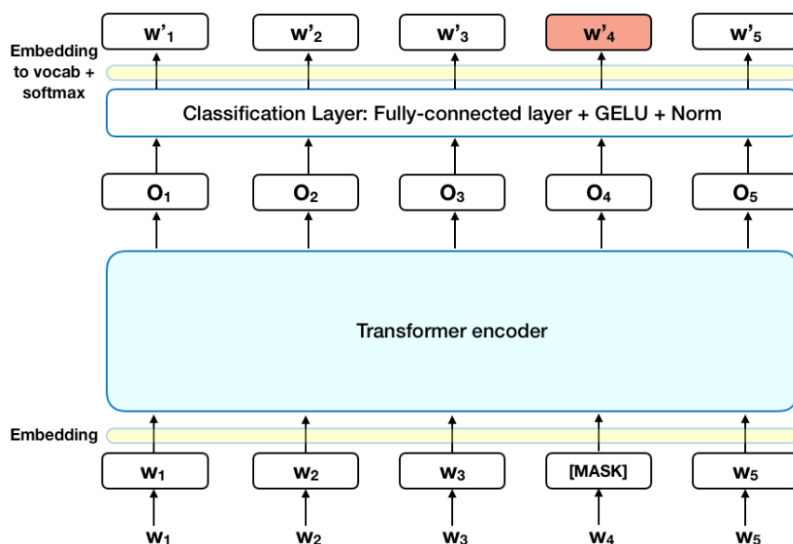
Original: ["Ion", "Popescu", "este"]

Labels: ["B-PERSON", "I-PERSON", "O"]

Tokenized: ["[CLS]", "Ion", "Pop", "##escu", "este", "[SEP]"]

Aligned: [-100, B-PERSON, I-PERSON, -100, O, -100]

This alignment ensures that only the first subword token of each word contributes to the NER loss, preventing the model from learning incorrect patterns from split tokens.



BERT Explained: State of the art language model for NLP

Before feeding word sequences into BERT, 15% of the words in each sequence are replaced with a [MASK] token. The model then attempts to predict the original value of the masked words, based on the context provided by the other, non-masked, words in the sequence. The BERT loss function takes into consideration only the prediction of the masked values and ignores the prediction of the non-masked words. As a consequence, the model converges slower than directional models, a characteristic which is offset by its increased context awareness.

Training Arguments

The fine-tuning process uses carefully selected hyperparameters optimized for domain adaptation:

****Learning Rate (3e-5)****: A relatively small learning rate is chosen to enable gradual adaptation. This prevents the model from "forgetting" its general Romanian language knowledge while learning legal domain specifics. The small learning rate ensures stable training and prevents catastrophic forgetting.

****Batch Size (16)****: The per-device batch size of 16 balances memory constraints with training stability. Larger batches provide more stable gradients but require more GPU memory. The batch size of 16 is a standard choice for BERT fine-tuning that works well across different hardware configurations.

****Number of Epochs (8)****: Eight training epochs provide sufficient opportunity for domain adaptation without overfitting. The model uses early stopping based on validation loss to prevent overtraining.

****Weight Decay (0.01)****: L2 regularization helps prevent overfitting by penalizing large weights. This is particularly important when fine-tuning on a relatively small domain-specific dataset.

****Warmup Ratio (0.1)****: The first 10% of training steps use a linearly increasing learning rate (warmup), gradually ramping up from zero to the full learning rate. This prevents early training instability that can occur when starting with a full learning rate.

****Evaluation Strategy****: The model evaluates on the validation set after each epoch (`'eval_strategy="epoch"'`), allowing monitoring of generalization performance throughout training.

****Model Selection****: The training process saves checkpoints after each epoch and automatically loads the best model based on validation loss (`'load_best_model_at_end=True', 'metric_for_best_model="eval_loss"'`). This ensures the final model represents the best generalization performance, not just the final training state.

Data Collator

The `DataCollatorForTokenClassification` handles dynamic batching:

- Padding: Pads sequences within a batch to the same length
- Attention Masks: Creates attention masks to indicate which tokens are real (not padding)
- Label Padding: Pads label sequences to match token sequences, using `-100` for padding positions

This dynamic padding maximizes efficiency by only padding to the longest sequence in each batch, rather than padding all sequences to the maximum length.

Fine-Tuning Process

The fine-tuning process follows a structured workflow:

1. Data Loading: Loads and processes the LegalNERo corpus
2. Model Initialization: Loads the pre-trained BERT model and configures the classification head
3. Tokenization: Applies tokenization and label alignment to all training and validation examples
4. Trainer Setup: Configures the Hugging Face Trainer with model, data, and training arguments
5. Training Loop: Executes the training process, which includes:
 - Forward pass: Model processes batches and generates predictions
 - Loss calculation: Cross-entropy loss between predictions and aligned labels
 - Backward pass: Gradients are computed and weights are updated
 - Validation: After each epoch, the model evaluates on the validation set
 - Checkpointing: Saves model checkpoints and selects the best model

Loss Function and Optimization

The training uses cross-entropy loss for multi-class token classification:

- Each token is classified into one of the NER label categories
- Tokens with label `-100` are ignored in loss calculation (subword tokens and special tokens)
- The loss is averaged over all valid tokens in the batch

The optimizer uses AdamW (Adam with weight decay), which is the standard optimizer for BERT fine-tuning. AdamW combines the adaptive learning rate of Adam with proper weight decay regularization.

This information is essential for:

- Understanding what entity types the model can recognize
- Properly interpreting model predictions
- Ensuring compatibility when loading the model in different contexts

The BERT architecture employed here consists of 12 transformer layers with 768 hidden dimensions, utilizing 110 million parameters. The model was fine-tuned for token classification using the BIO (Beginning-Inside-Outside) tagging scheme, where each token is labeled as either:

- `**B-{LABEL}**`: Beginning of an entity

- **I-{LABEL}**: Inside/continuation of an entity
- **O**: Outside/not part of any entity

The fine-tuning process adapted the pre-trained BERT weights to recognize 15 distinct entity types relevant to Romanian legal documents:

- **PERSON**: Person names (e.g., "Ion Popescu", "Maria Ionescu")
- **ORG**: Organizations (e.g., "Google", "Primăria București")
- **LOC**: Locations (e.g., "București", "Cluj-Napoca")
- **GPE**: Geopolitical entities (e.g., "România", "Uniunea Europeană")
- **DATETIME**: Specific dates and times (e.g., "15 ianuarie 2020")
- **PERIOD**: Time periods (e.g., "2020-2021", "anii 90")
- **MONEY**: Monetary amounts (e.g., "5000 RON", "7500 EUR")
- **FACILITY**: Facilities and buildings
- **EVENT**: Events
- **LANGUAGE**: Languages
- **WORK_OF_ART**: Artistic works
- **NAT_REL_POL**: National, religious, and political entities
- **NUMERIC**: Numeric values
- **ORDINAL**: Ordinal numbers
- **QUANTITY**: Quantities

To enable high-performance inference in a Rust-based production environment, the PyTorch model is converted to ONNX (Open Neural Network Exchange) format. This conversion process, facilitated by HuggingFace's `optimum` library, transforms the model into a standardized format that can be executed efficiently by ONNX Runtime without requiring the full PyTorch framework.

The conversion process preserves the model's architecture and weights while optimizing the computational graph for inference. The resulting ONNX model (approximately 400-500 MB) includes all necessary operations for token classification, including the embedding layers, transformer blocks, and the final classification head that outputs logits for each token and label combination.

The proliferation of digital documents containing sensitive personal information has created an urgent need for automated anonymization systems. In Romania, where legal and administrative documents frequently contain names, identification numbers, financial information, and other personally identifiable data, manual redaction is both time-consuming and error-prone. This project addresses this challenge through a sophisticated two-stage approach: first, employing a fine-tuned BERT model for contextual entity recognition, and second, applying rule-based patterns for structured PII detection that may not be captured by the neural model alone.

The system's architecture demonstrates a pragmatic balance between the flexibility of deep learning and the precision of rule-based systems. By combining these approaches, we achieve high recall for named entities while maintaining strict accuracy for structured identifiers like Romanian CNP (Cod Numeric Personal) numbers and IBAN bank account codes, which require format validation beyond simple pattern matching.

While the BERT model excels at recognizing contextual entities like person names and organizations, structured identifiers require precise pattern matching and validation. The system implements rule-based detection for several PII categories that benefit from format validation:

Rule-Based PII Detection: Complementing Neural Recognition

Romanian CNP Detection

The Romanian Cod Numeric Personal (CNP) is a 13-digit personal identification number with a specific structure:

- First digit (1-8): Gender and century of birth
- Digits 2-7: Date of birth (YYMMDD)
- Digits 8-9: County code
- Digits 10-12: Sequential number
- Digit 13: Check digit

The system uses a two-stage validation approach:

1. **Pattern Matching**: Regex pattern `\b[1-8]\d{12}\b` identifies 13-digit sequences starting with 1-8
2. **Checksum Validation**: The MOD 11 algorithm validates the check digit using the formula:

$$\text{sum} = \sum (\text{digit}[i] \times \text{control}[i]) \text{ for } i \text{ in } 0..11$$

$$\text{check} = (\text{sum} \% 11 == 10) ? 1 : (\text{sum} \% 11)$$

where `control = [2, 7, 9, 1, 4, 6, 3, 5, 8, 2, 7, 9]`

IBAN Detection and Validation

International Bank Account Numbers (IBANs) follow a standardized format:

- 2 letters: Country code (e.g., "RO" for Romania)
- 2 digits: Check digits
- Up to 30 alphanumeric characters: Bank and account identifier

The system validates IBANs using the MOD 97-10 algorithm:

1. Move the first 4 characters to the end
2. Convert letters to numbers (A=10, B=11, ..., Z=35)
3. Calculate `remainder = large_number % 97`
4. Valid IBANs have `remainder == 1`

This validation prevents false positives from random alphanumeric sequences that happen to match the IBAN pattern.

Phone Number Detection

Romanian phone numbers exhibit multiple formats:

- **Local format**: 10 digits, optionally starting with "0" (e.g., "0712345678")
- **International with country code**: "+40" followed by 9-10 digits (e.g., "+40712345678")
- **International with 00 prefix**: "00" followed by country code and number (e.g., "0040712345678")

The detection regex accommodates these variations while filtering out sequences that are actually CNPs (13 digits) or parts of IBANs.

Email Address Detection

Email addresses are detected using a standard regex pattern that matches:

- Local part: Alphanumeric characters, dots, plus signs, hyphens
- "@" symbol
- Domain part: Alphanumeric characters, dots, hyphens
- Top-level domain: 2+ letters

Entity Mappings

The system generates a comprehensive mapping file that records all anonymized entities:

[PERSON1] = Ion Popescu, ION POPESCU

[PERSON2] = Maria Ionescu

[ORG1] = Google

[CNP1] = 1234567890123

[IBAN1] = RO49AAAA1B31007593840000

[EMAIL1] = ion@gmail.com

[PHONE1] = 0712345678

This mapping enables:

- Audit trails: Tracking what was anonymized
- Re-identification: If authorized, entities can be restored
- Quality assurance: Verifying that anonymization was correct

This Romanian NER and PII anonymization system demonstrates the effectiveness of combining deep learning with rule-based methods. The fine-tuned BERT model provides contextual understanding for named entities, while rule-based detection ensures precision for structured identifiers requiring format validation. The sophisticated deduplication and numbering system ensures consistent anonymization throughout documents, while the dual output modes (standard and date-preserving) accommodate different use cases.

The system's architecture balances accuracy, performance, and maintainability. By leveraging ONNX for model deployment, the system achieves production-grade performance while remaining accessible through a simple command-line interface. The comprehensive entity mapping output provides transparency and enables quality assurance, critical for legal and administrative applications.

The system represents a practical solution to a real-world problem, demonstrating how modern NLP techniques can be applied to privacy-preserving document processing in specialized domains.

Original:

```
CONTRACT DE TRANZACȚIE
din data de [...]
```

Acest contract de tranzacție („Tranzacția” sau „Contractul”) este încheiat în baza art. 2267 și următoarele din Cod Civil de către:

1. BOBU MARIUS-IRINEL, născut la data 10.02.1974, în Pitești, jud. Argeș, cu domiciliul în Jud. Argeș, Mun. Pitești, str. Eugen Ion
2. BOBU CLAUDIA, născută în Pitești, jud. Argeș, la data de 19.01.1979, cu domiciliul în Jud. Argeș, Mun. Pitești, str. Eugen Iones
3. EL-SIBLANI FAOUZI-DANIEL, născut în Pitești, jud. Argeș, la data de 18.03.1987, cu domiciliul în Jud. Argeș, Mun. Pitești, Cal. Denumiți în mod individual „Asociatul A” și în mod colectiv „Asociații A”,
4. GHEORGHE NICOLETA, născută în Cugir, jud. Alba, la data de 16.07.1986, cu domiciliul în Jud. Argeș, sat. Smeura (Com. Moșoaia), Denumită, în continuare, „dna Gheorghe Nicoleta”.

Asociatul A și dna Gheorghe Nicoleta, vor fi denumiți în continuare, în mod individual „Asociatul” și colectiv „Asociații”, după caz

5. FMNC LUX S.R.L., societate înmatriculată și funcționând în conformitate cu legea română, având sediul social în Mun. Pitești, B-

Art. 1. PREAMBUL

- 1.1. Asociații dețin, în Societatea de tip Holding denumită FMNC LUX, următoarele proporții:
- 1.1.1. Bobu Marius-Irinel deține 20% din capitalul social al Societății, reprezentând un număr de 4 părți sociale, cu o valoare nom
- 1.1.2. Bobu Claudia deține 20% din capitalul social al Societății, reprezentând un număr de 4 părți sociale, cu o valoare nominală
- 1.1.3. El-Siblanî Faouzi-Daniel deține 30% din capitalul social al Societății, reprezentând un număr de 6 părți sociale, cu o valo
- 1.1.4. Gheorghe Nicoleta deține 30% din capitalul social al Societății, reprezentând un număr de 6 părți sociale, cu o valoare nomi
- 1.2. Părțile au încheiat și semnat Acordul Asociaților din data de 23.01.2024, ulterior completat prin Actul Adițional la Acordul

a) Sistemul Epiclinique - dezvoltat de Bobu Marius-Irinel și Bobu Claudia sub marca „Epiclinique”, aceștia deținând, atât împreună

Result:

```
CONTRACT DE TRANZACȚIE
din data de [...]
```

Acest contract de tranzacție („Tranzacția” sau „Contractul”) este încheiat în baza art. 2267 și următoarele din Cod Civil de către:

1. [PERSON1], născut la data 10.02.1974, în [LOC1], jud. Argeș, cu domiciliul în Jud. [LOC2], Mun. [LOC1], str.[LOC3], bl. A13, sc.
2. [PERSON2], născută în [LOC1], jud. [LOC2], la data de 19.01.1979, cu domiciliul în Jud. [LOC2], Mun. [LOC1], str. [LOC4] nr. [L
3. [PERSON3], născut în [LOC1], jud. [LOC2], la data de 18.03.1987, cu domiciliul în Jud. [LOC2], Mun. [LOC1], Cal. [LOC6] nr. 13, Denumiți în mod individual „Asociatul A” și în mod colectiv „Asociații A”,
4. [PERSON4], născută în [LOC7], jud. Alba, la data de 16.07.1986, cu domiciliul în Jud. [LOC2], sat. [LOC8] (Com. [LOC9]), Ale. [Denumită, în continuare, „dna [PERSON4]”.

Asociatul A și dna [PERSON4], vor fi denumiți în continuare, în mod individual „Asociatul” și colectiv „Asociații”, după caz, în mo

5. [ORG2]., societate înmatriculată și funcționând în conformitate cu legea română, având sediul social în [LOC12], [LOC13], [LOC1

Art. 1. PREAMBUL

- 1.1. Asociații dețin, în Societatea de tip Holding denumită [ORG5], următoarele proporții:
- 1.1.1. [PERSON1] deține 20% din capitalul social al Societății, reprezentând un număr de 4 părți sociale, cu o valoare nominală de
- 1.1.2. [PERSON2] deține 20% din capitalul social al Societății, reprezentând un număr de 4 părți sociale, cu o valoare nominală de
- 1.1.3. [PERSON5] deține 30% din capitalul social al Societății, reprezentând un număr de 6 părți sociale, cu o valoare nominală de

Expected Outcomes

The fine-tuning process produces a model capable of:

- Recognizing Legal Entities: Accurately identifies persons, organizations, locations, and time expressions in legal documents
- Legal Document Classification: Identifies and classifies various types of legal references (laws, ordinances, decisions, etc.)
- Domain Adaptation: Understands legal terminology and context better than the general-purpose base model
- Romanian Language Support: Maintains full support for Romanian language features, including diacritics and complex grammatical structures

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